

Neonatal and Pediatrics Anesthesia

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Part two

ANESTHESIA FOR SPECIFIC NEONATAL SURGICAL PROCEDURES

CONGENITAL HEART DISEASE (CHD)

- Diverse anatomic defects and physiologic changes
- Altered ventricular remodeling owing to myocardial hypertrophy and ischemia
- Chronic sequelae of congenital cardiac disease

CONT'D

- **Physioanatomic** difference from acquired adult cardiac disease
- Presence of intracardiac shunts, valve pathologies, difects, and absence of one or more heart chambers
- Preclude a **uniform anesthetic** approach to this patients
- Functional myocardial changes may lead to intraop ventricular ischemia and failure

CONT'D

- Understanding the isolated defect, associated myocardial changes, and hemodynamic consequences is important for planning an appropriate anesthetic regimen
- To have a predictable impact of drugs, ventilatory management, and fluid administration to optimize cardiovascular performance

A general physiologic classification

- **Left-to-right shunts**
 - VSD, ASD, PDA and AV canal
- ↑ pulmonary BF
- Volume-overloaded ventricle, Development of **acyanotic** CHF
- **Right-to-left shunts**
 - Tetralogy of Fallot (TOF), Pulmonary atresia/VSD
- ↓ Pulmonary BF, Pressure-overloaded ventricle, **Cyanotic** CHD

CONT'D

- **Mixing lesions**

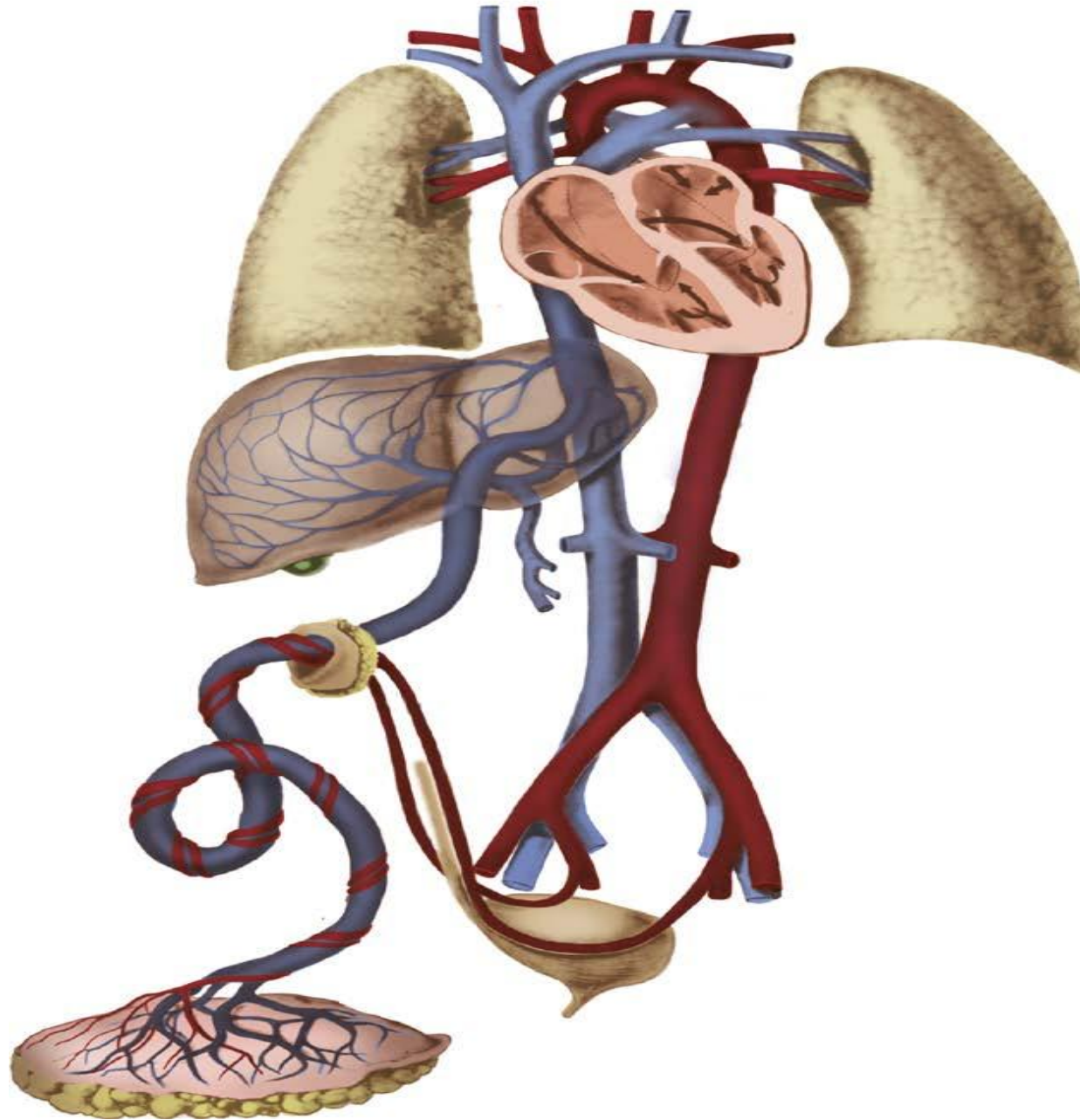
- Transposition/large VSD
 - Tricuspid atresia
 - Total anomalous pulmonary venous return, pulmonary atresia
- ↓ Pulmonary BF
 - Variable pressure versus volume loaded
 - Usually cyanotic

CONT'D

- **Obstructive lesions**
 - Interrupted aortic arch
 - Critical aortic stenosis
 - Critical pulmonic stenosis
 - Hypoplastic left heart syndrome (HLHS)
 - Coarctation of the aorta
 - Mitral stenosis
- Ventricular dysfunction
- Pressure-overloaded ventricle
- Ductal dependence

CONT'D

- **Regurgitant lesions**
 - Ebstein anomaly (Tricuspid valve malformation)
 - Other secondary causes
- Volume-overloaded ventricle
- Development of CHF



CONT'D

- When considering the incidence of all the CHDs,
- Three left-to-right shunts (VSD, ASD, PDA) and
- Two obstructive lesions (pulmonic stenosis, aortic coarctation) constitute 60% of all congenital cardiac defects
- Mixing lesions, complicated obstructive defects, and right-to-left shunting lesions accounts for the remaining 40% of defects

ANESTHETIC MANAGEMENT

PREOPERATIVE MANAGEMENT

- **Anesthetic Evaluation**
- Range from a small ASD to a neonate with HLHS that needs aggressive perioperative hemodynamic and ventilatory support
- Family may to meet the surgeon and anesthesiologist (detail discussion)
- Obtain previous, medical/surgical hx, current medications, prior anesthetic problems, and family hx of anesthetic difficulties
- Echo, and cardiac catheterization, physical examination rarely needed, but indicative of the prognosis.

Concurrent Medications and Drug Interactions

- **Angiotensin-converting enzyme inhibitors**
- Hypotension with induction of general anesthesia
- **Remedy-withhold morning dose**, or reducing dosage, in patients with hypotension; avoid fixed dose induction regimens with drugs having a **profound vagomimetic** effect

CONT'D

- **β-Blockers**
- Acute withdrawal can precipitate tachycardia and arrhythmias;
- Can potentiate hypotension with IAAs
- Can decrease response to inotropic agents
- **Remedy-** Continue in the perioperative period

CONT'D

- **Calcium channel blockers**
- May augment the negative inotropic and chronotropic effects of volatile anesthesia
- **Remedy**-Continue in the perioperative period

CONT'D

- **Diuretics**
- Hypovolemia/hypokalemia
- May augment effect of neuromuscular blocking agents
- Remedy-Discontinue preoperatively

CONT'D

- **Antiarrhythmics**
- Can be proarrhythmic with inotropes
- Electrolyte disturbances
- High catecholaminergic states
- Interacts with other antiarrhythmics and precipitate bradycardia
- **Remedy-**
- Avoid electrolyte imbalance
- Avoid drugs that are proarrhythmic
- Monitor carefully

CONT'D

- **α 2-Agonists**
- Reduces perioperative shivering
- Reduce Ischemia
- Reduce anesthetic and analgesic requirements
- **Remedy-**
- Continue perioperatively with appropriate monitoring

CONT'D

- Prolonged high ($>60\%$) Hct may show chronic hypoxia
- Echo-Doppler is helpful to identify intracardiac abnormalities
- CT, and cardiac catheterization is helpful to identify extracardiac abnormalities (pulmonary artery or venous stenosis)
- Cardiac catheterization remains the **gold standard** for assessing anatomy and physiologic function in CHD

INTRAOPERATIVE MANAGEMENT

- A careful OR preparation, anesthesia machine, with capacity to provide air, O₂, and N₂O (balance PV and SV flow), monitors
- Avoid air bubbles to prevent **paradoxical air embolism**
- **Resuscitative drugs** like Succinylcholine, calcium gluconate, sodium bicarbonate, atropine, lidocaine, phenylephrine, and epinephrine be labeled and ready for administration

CONT'D

- Prepare **Etomidate**, propofol, ketamine
- These children may have limited reserve and high endogenous catecholamine released in an adaptive response to their cardiac disease
- The resuscitative drugs should be prepared and immediately available before anesthetic induction

Monitoring of Organ Systems

- **Cardiopulmonary System**

- Esophageal stethoscope, ECG, Pulse oximetry, BP, Capnograph,
Indwelling arterial catheter, CVP or PA catheter, pulmonary artery
Echo with Doppler color flow imaging

CONT'D

- **Central Nervous System**
- Peripheral nerve stimulator, Processed EEG,
- Specialized
 - CBF :Xenon clearance methodology
 - Cerebral metabolism:Near-infrared spectroscopy, oxygen consumption measurements
 - Transcranial Doppler
 - Jugular venous bulb saturations

CONT'D

- **Temperature**
- Nasopharyngeal, rectal, esophageal, tympanic
- **Renal Function**
- Foley catheter
- **Glucose**
- Frequent glucometry measurement

Induction and Maintenance of Anesthesia

- Anesthesia induction technique depends the degree of cardiac dysfunction, the cardiac defect, the degree of sedation provided by the **premedication, and the presence of an indwelling catheter**
- For those with good cardiac reserve, induction can be quite varied in well-monitored patients
- **Titration of anesthesia induction** is more important than the specific anesthetic technique

CONT'D

- IV and inhalational induction have been used safely
- **Ketamin** induction is used in patients with cyanotic conditions (**it↑es SVR and CO, ↓sing right-to-left shunting**)
- IV induction will be faster in right -to- left shunt
- Inhalational induction will be delayed in left-to-right shunt

CONT'D

- An inhaled induction of anesthesia with sevoflurane can be safely be performed even in cyanotic (TOF) patients
- For those at risk of right-to-left shunting and systemic desaturation, oxygenation is well maintained with a good airway and ventilation despite reduction in SVR
- **Skilled AW mgt** and efficiency of ventilation are equally essential components of anesthetic induction

CONT'D

- A NDMR is given and an IV opioid and/or inhaled anesthetic are chosen for maintenance of anesthesia
- After preoxygenation with 100% FiO₂ place ETT carefully
- Some degree of preoxygenation is required, in those systemic perfusion is affected lowering PVR with resulting increase in PBF, as this maneuver delays desaturation during intubation

CONT'D

- An individualized anesthetic mgt plan is essential
- Anesthesia maintenance depends on the:
 - Age and condition of the patient
 - The nature of the surgical procedure
 - The duration of the CPB and
 - Need for postoperative ventilation
- Hemodynamic assessment is designed to lessen pathophysiologic loading conditions should be developed for each patient

CONT'D

- In patients with complex defects requiring preoperative inotropic and mechanical ventilation:
- A carefully controlled induction and maintenance anesthetic with a potent opioid is generally chosen
- In patients with a simple ASD or VSD:
- A balanced anesthetic with IAAs, opioid titration, and the use of dexmedetomidine is preferred

CONT'D

- Despite improved cardiac reserve with isoflurane:
 - The incidence of laryngospasm
 - Coughing
 - Desaturation during induction limits its induction use
 - Desflurane has cardiorespiratory properties similar to isoflurane
 - Its main advantage is **low blood gas and tissue solubility**

CONT'D

- This enables for greater precision in drug dosing during the operative period and may make desflurane a more titratable and preferred for pediatric cardiac anesthesia
- The three main disadvantages of desflurane are:
- Potency, pungency, and negative inotropic effect
- It should not be used as the sole anesthetic in patients with significant cardiac dysfunction

CONT'D

- **Sevoflurane**, our volatile anesthetic of choice, offers a more tolerable condition than others
- Reductions in HR and SVR are more modest in infants anesthetized with sevoflurane
- **Fentanyl** is an excellent induction and maintenance adjunct in these patients
- Low-to-moderate doses of this opioid can be supplemented with inhalation anesthetics

CONT'D

- **Remifentanyl**, an ultra–short-acting opioid, offers the unique advantage of metabolism by **nonspecific and tissue esterases**, thereby limiting the potential for accumulation related to protracted elimination
- Intraoperative use of **methadone** is an alternative pain control strategy introduced as an answer to counter acute tolerance to fentanyl infusions

CONT'D

- Dexmedetomidine is an α_2 -agonist approved by the U.S FDA for sedation in adults
- **Preoperatively** it is used as part of a balanced technique and
- **Intraoperatively** for sedation, anxiolysis, and analgesia, and
- **Postoperatively** for prevention of emergence delirium and sedation

CONT'D

- **Dexmedetomidine** has significant analgesic and antiinflammatory effects attenuates the neuroendocrine response to surgery, and
- It has no neurotoxic effects; it is a crucial adjunct to a balanced anesthetic as it reduces the need for other analgesics and hypnotics

Hyperthrophic Pyloric Stenosis (HPS)

Pathophysiology HPS

- Relatively frequent surgical disease of neonate and infant
- **Etiology** : unknown ; acquired condition with hereditary predisposition
- The pathologic characteristics include:
 - Hypertrophy of the pyloric smooth muscle with
 - Edema of the pyloric mucosa and submucosa

CONT'D

- This develops over days to weeks, leads to progressive obstruction of the pyloric valve, causing persistent vomiting
- That leads to loss of fluids and electrolytes
- This vomiting depletes Na^+ , K^+ , Cl^- and H^+ ions
- Hypochloremic hypokalemic metabolic alkalosis

CONT'D

- The kidney tries to compensate for the alkalosis (excretion of sodium bicarbonate)
- Later, hyponatremia and dehydration worsen
- The kidneys must conserve sodium (at expense of H-excretion, paradoxical aciduria) Correction of this deficit requires hydration with a sodium chloride solution supplemented with potassium
- Lactated Ringer's injection should not be used (why??)

CONT'D

- This procedure is never a surgical emergency (medical instead)
- Symptoms are apparent b/n 2nd -6th week of life
- Presents with :
 - Nonbilious projectile vomiting
 - Signs of dehydration, jaundice (2%)

Diagnosis

- ❖ History and physical exam
- ❖ abdominal ultrasound
- ❖ upper GI series with barium contrast not recommended
- ❖ pathological
 - ✓ pyloric wall thickness ≥ 4 mm
 - ✓ pyloric length of > 16 cm
- ❖ Physical Exam

CONT'D

- ✓ visible gastric peristalsis
- ✓ palpable “olive-shaped” mass to right of epigastric area
- ✓ signs of dehydration
- ❖ Labs: CBC, serum electrolytes, ECG, ABG, BUN



Metabolic Abnormalities

- Hyponatremia
- Hypochloremia
- Hypokalemia
- 1^o Metabolic alkalosis
- Compensatory respiratory acidosis
- Paradoxical aciduria

Preoperative Preparation

- Supportive treatment
- Surgical mgt
- Check lab indices for safe anesthesia

Supportive therapy

- Correction of fluid deficits
maintenance: D5 0.2% NaCl + KCl = 20 - 40 mEq/L
replacement: , albumin, normal saline, LR?
- Correction of electrolyte imbalance
- Prevention of aspiration : **NGT**

CONT'D

Surgical Management

Pyloromyotomy

definitive treatment

open or laparoscopic

Lab indices for safe anesthesia

- serum Cl $>85-100$ mEq/L
- $\text{HCO}_3^- < 28$ mEq/L
- Sodium >130 mEq/L
- potassium at least 3 mEq/L
- urine output of at least 1 to 2 mL/kg/hr.

Anesthetic Concerns

- Severe dehydration
- Metabolic alkalosis
- Respiratory acidosis
- Monitor: ASA Standards of monitoring
- Decompress the stomach

CONT'D

- Techniques of intubation and induction vary
- Consider high risk of aspiration
- Experts advocate awake intubation, RSI?, or
- A careful inhalation induction in selected patients
- Pyloromyotomy is a short procedure (needs muscle relaxation)

CONT'D

- **Maintenance**
- IV narcotics :rarely needed
- Inhalational agents

Postop pain relief

- Acetaminophen 30-40 mg/kg PR,
- Caudal epidural
- LA infiltration of surgical incision
- Extubate awake

Postoperative concerns

- respiratory depression and apnea
- CFS alkalosis

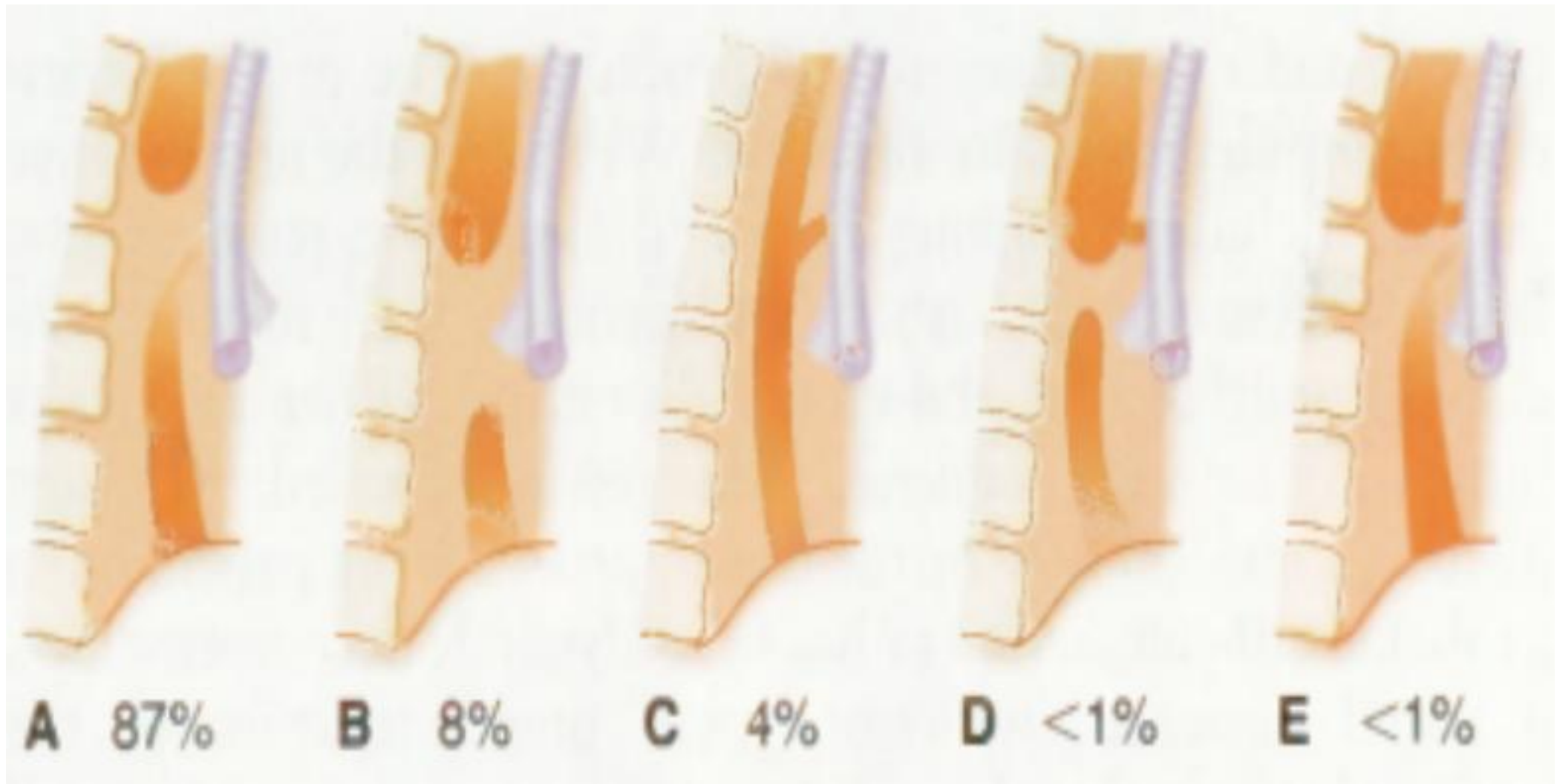
Introduction to TEF and EA

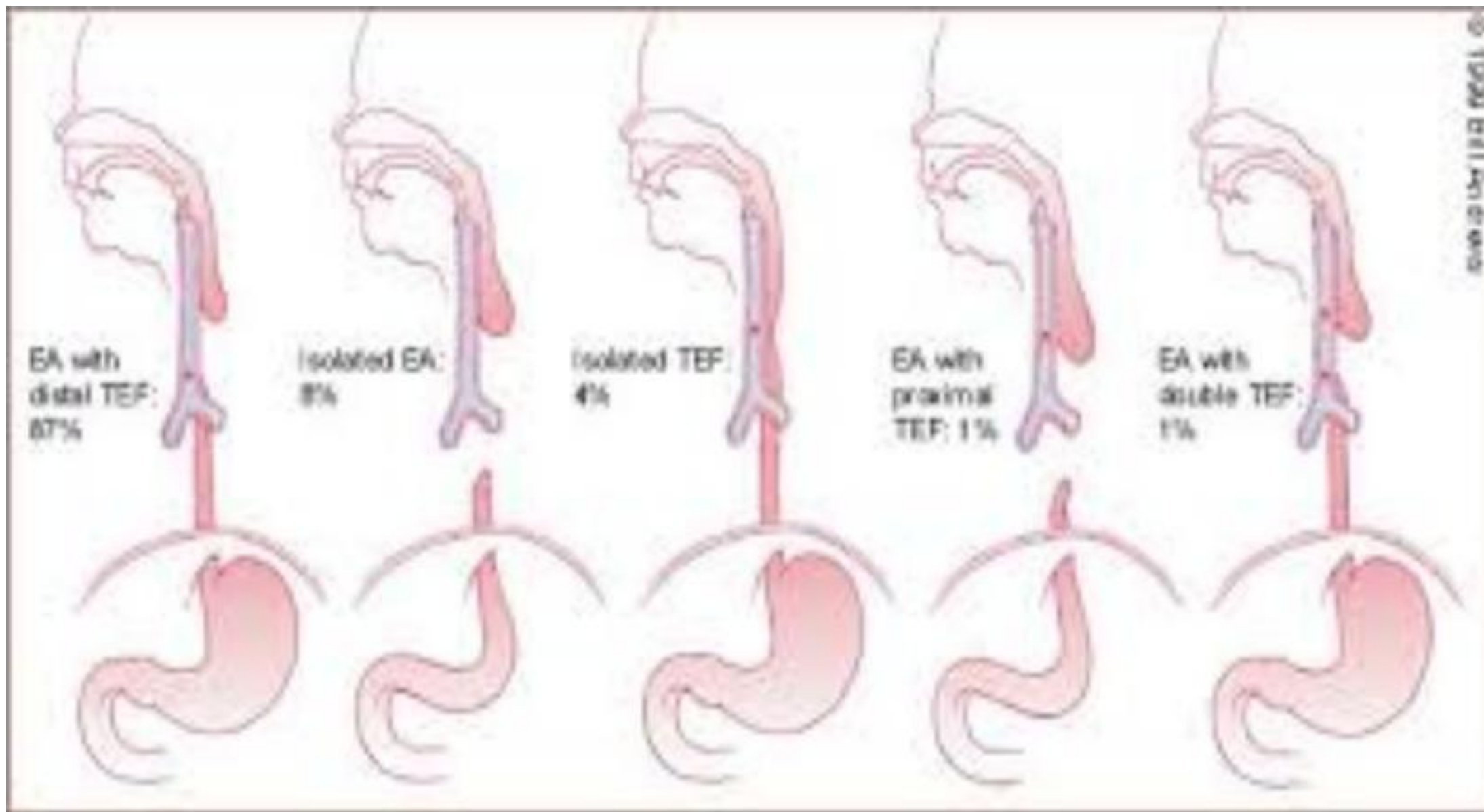
Epidimology

- Due to failed fusion of TE ridges during 3rd wk of embryological life
- Incidence of 1 in 4500 live births (USA data)
- Relatively common congenital anomalies
- 0.5% - 2% ↑se risk in newborns with one affected sibling
- Risk ↑ses to 20% if more than one sibling is affected

Anatomic Characteristics

- Esophageal atresia with distal TEF-87%, the most common
- Isolated esophageal atresia without TEF-8%
- Isolated TEF -4%
- Esophageal atresia with proximal TEF -1%
- Esophageal atresia with proximal and distal TEF -1%





CONT'D

Associated Anomalies VACTERL:



➤ Congenital heart disease is associated with higher mortality

- VSD is most common
- ASD
- Tetralogy of Fallot
- PDA

➤ ECG – important to determine position of aortic arch

V- vertebral anomalies or VSD

A -anorectal malformation

C -cardiac anomalies (common)

T -TEF

E -esophageal atresia

R -renal abnormalities

L -limb/radial malformation

Clinical Features

- **Prenatally**
 - Predictive value of prenatal ultrasound ~20-40%
 - Polyhydramnios (2/3 cases)
 - Fetus unable to swallow amniotic fluid
 - Small or absent stomach bubble

CONT'D

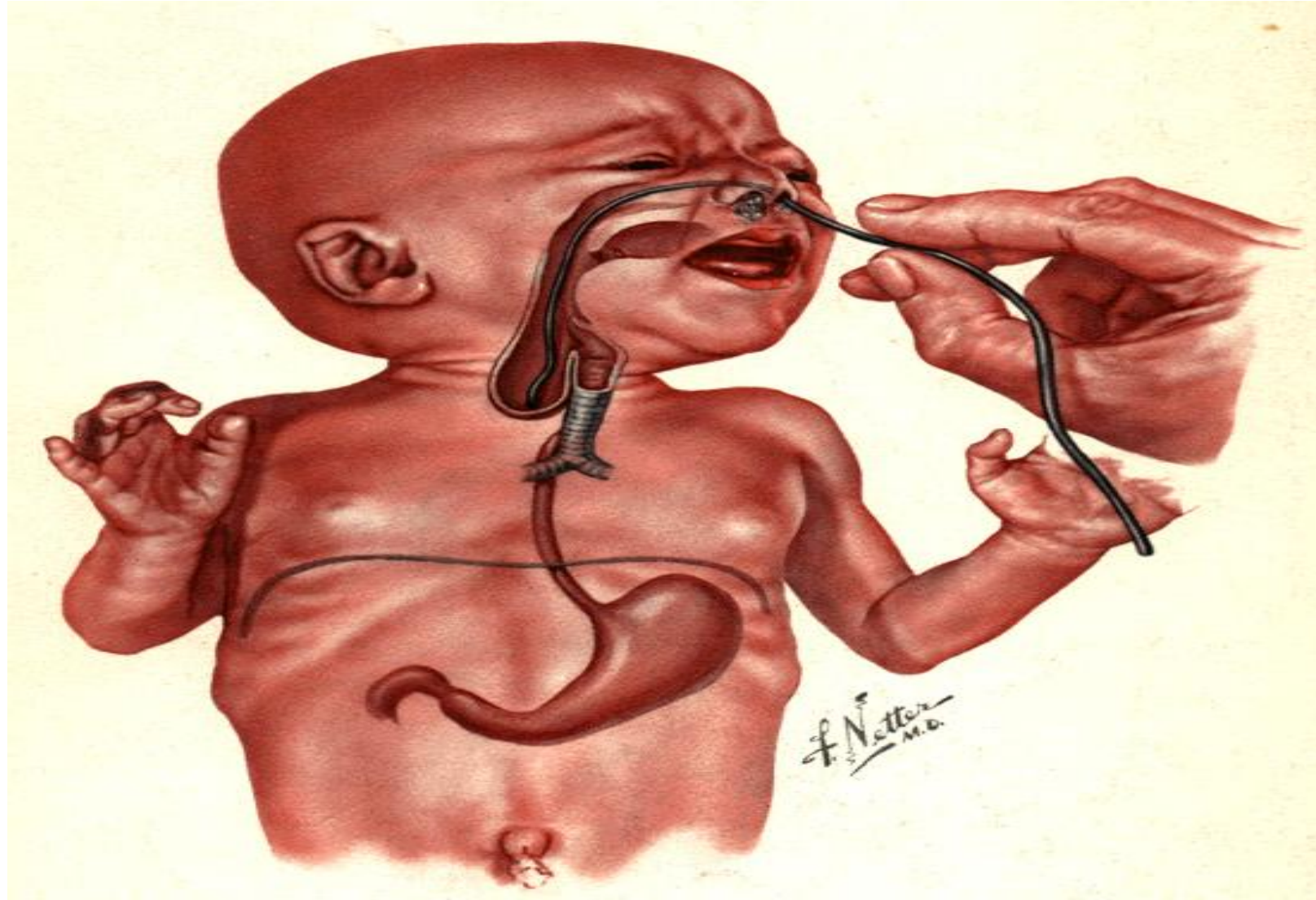
- **Postnatally**

- Most infants are symptomatic within first few hours of life
- Excessive salivation
- Regurgitation of first feed
- Cyanosis with/without feeds
- Respiratory distress
- Inability to pass a catheter into the stomach
- Gastric distention (with distal fistula)

Diagnosis

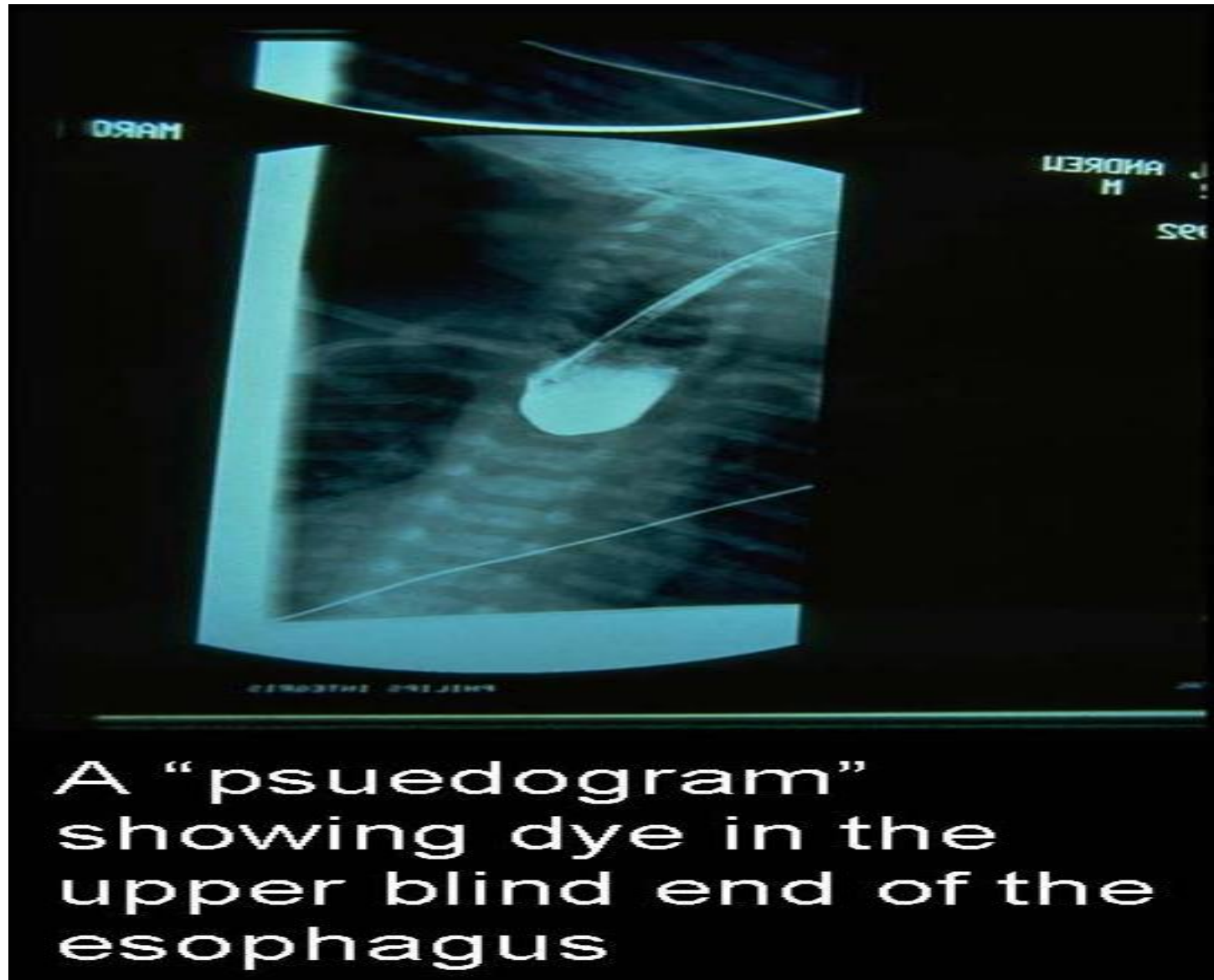
- Breathing results in gastric distention..???
- Feeding leads to **choking, coughing, and cyanosis** (3 C's)
- Failure to pass NG tube
- CXR- tube coiled in upper esophagus
- “Pouchogram” with air
- Distal air on AXR confirms the presence of a fistula
- Pass catheter into stomach. Cannot pass more than 10-15 cm

CONT'D



CONT'D

- If still diagnosis is uncertain or proximal TEF is suspected, a small amount of **water-soluble contrast** material can be injected into the esophageal pouch under fluoroscopic guidance
- The contrast must be removed immediately to avoid aspiration and regurgitation



Surgical Management

- Surgical ligation of the fistula
- Primary anastomosis of the esophageal segments
- Primary repair may not be possible if the distance between esophageal segments is long
- Staged procedures have been performed that include elongation of the esophagus with circular myotomy, interposition of the colon, and gastric transposition

Anesthetic Management

- Preoperatively

- Focus on associated CHD, impediments to pulmonary compliance, other co-morbid disease
- Consider the possibility of concomitant airway anomalies
- NPO/TPN..
- Upright position of child
- Early surgery for short gap atresia
- Long gap atresia may be delayed up to 6-12 weeks
- Long gap atresia should have **gastrostomy** tube placed for enteral nutrition

CONT'D

- The major anesthetic issues include:
 - An already compromised RS (aspiration pneumonia)
 - Overdistention of the stomach (direct air entry mainly PPV by mask)
 - Inability to ventilate the child's lungs (large sized fistula)
 - Problems associated with other anomalies(PDA, other CHD)

Induction and Intraoperative

- Minimize complications from aspiration
- Suction blind pouch continuously
- Effective communication is important
- Catastrophic risk of tube malposition
 - Distal- minimizes gastric insufflations and improves both ventilation and the surgical field
 - Proximal to fistula - avoids malposition and no disturbance to mechanical ventilation

CONT'D

- Risk of intubating the fistula-difficulty in ventilation ,decreased SaO₂ and EtCO₂.
- Invasive monitoring, BP and SpO₂
- Gastrostomy reduces the potential for reflux of gastric juice during surgery

Three tracheal intubation approaches

- **Inhalation induction**, followed by topical spray of lidocaine and intubation while the infant is breathing spontaneously.
- IV or inhalation induction and intubate the trachea after muscle paralysis-**distention of the fistula & stomach with excessive PPV.**
- Awake intubation with mild sedation-can protect airway & reduce the chances of aspiration

The endotracheal tube can be placed just distal to the TEF and proximal to the carina.

Extubation and Post Operative Management

- Extubation may be complicated by poor pain control, tracheomalacia and vocal cord paralysis
 - Dysphagia, GERD and bronchial hypereactivity

Post operative care

- Adequate fluid resuscitation
- Wean ventilator
- No deep tracheal suctioning
- Avoid bag/mask ventilation
- No pacifiers/sucking
- Chest tube remains in place for approximately one week
- Esophagram 1 week post op

PROGNOSIS

- 85% to 95% survival
- Infants with increased risk of morbidity and mortality
 - Low birth weight (>1500)
 - Major CHD
 - Severe associated anomalies
 - Ventilator dependency
 - Long gap atresia

Outcome

- **Complications**

- Anastomotic Leak
- Esophageal stricture
- Recurrent fistula

- **Other Associated Issues**

- Tracheomalacia
- Disturbed peristalsis and delayed gastric emptying are common
- “Seal Bark” cough
- Gastroesophageal reflux and aspiration

Congenital Diaphragmatic Hernia (CDH)

Introduction

- At fetal development, gut can herniate into one of three possible diaphragmatic defects:
 - The left posterolateral foramen of Bochdalek (90 %)
 - The right posterolateral foramen of Bochdalek or
 - The anterior foramen of Morgagni
- The incidence is reported to be 1 in 3000-5000 live births
- Hypoxia, a scaphoid abdomen, and evidence of bowel in the thorax by auscultation or radiography are hallmarks of diaphragmatic herniation

CONT'D

- Etiology
 - Cause of CDH is unknown
 - Increasing evidence that it may be related to exposure to environmental and genetic factors

CONT'D

- Associated Anomalies
 - Incidence ranges from 10-50%
 - Skeletal defects ~32%
 - Cardiac anomalies ~24%
 - Tracheobronchial tree ~18%

Pathology

- In most cases defect is established by 12 week GA
- Herniated contents are left lobe of the liver, the spleen, and almost the entire GI tract
- Persistent compression of developing fetal lung hypoplasia
- All the major bronchial buds are present in the CDH lung
- The number of bronchial branches is greatly reduced
- Alveolar(functional unit) development is also severely affected

CONT'D

- Classic Triad

- Dyspnea
- Cyanosis
- Apparent dextrocardia

- Pulmonary vascular bed is abnormal, resulting in increased pulmonary vascular resistance
- Increasing hypoxia results in increasing right-to left shunting leading to severe and progressive respiratory failure.

Diagnosis and Physical exam

- Prenatal u/s is accurate in 40-90% of cases
- Mean GA is 24 wks, some may be seen at 11 weeks of GA
- After birth the respiratory symptoms are determined by the degree of **pulmonary hypoplasia** and reactive **pulmonary hypertension**
- Most severely affected newborns develop respiratory **distress at birth**
- Majority shows symptoms within **the first 24 hours**

Hallmarks of CDH

- Hypoxia (Pulmonary hypoplasia)
- Scaphoid abdomen and symmetrically distended chest
- Bowel in thorax (auscultation, x-ray)
- Intestinal mal-rotation
- Pulmonary hypertension (40-50%)
- Polyhydramnios - in up to 80% of cases
- Tracheal deviation away from the defect

Physical Exam

- Scaphoid abdomen
- Barrel chest
- Bowel sounds in the chest
- Displaced heart sounds

LAB Ixs

- Laboratory Studies

CBC

ABG

electrolytes

calcium

glucose

Sodium

Anesthesia Management

- CDH is a **physiologic emergency NOT a surgical emergency**
- Early surgery further decreases lung compliance
- Transition of fetal circulation for vascular and pulmonary function is prolonged in CDH
- In theory, delayed surgery provides additional time for this transition to occur resulting in a more stable infant

CONT'D

- Ventilation by mask or Ambu bag is contraindicated
- May be managed on a conventional ventilator or a high frequency ventilator
- Key to success is currently thought to be gentle ventilation with permissive hypercapnea to reduce barotrauma

CONT'D

- Preductal pulseoximetry should be monitored
- Metabolic acid base disturbances should be corrected with fluid management or bicarbonate administration
- ECMO in severe cases - but often NOT necessary

Goals of Management

- Maximize arterial oxygenation

mechanical ventilation: use low inflating pressures

increases pulmonary blood flow

- Prevention of pain
 - Fentanyl infusion 3-10 mcg/kg/hr
 - Local anesthesia infiltration or blokcs
- Correction of acidosis

Standard Management Strategy

- Reduce pulmonary HTN
- Moderate alkalosis required
- $\text{paCO}_2 < 40 \text{ mmHg}$
- $\text{PaO}_2 > 100 \text{ mmHg}$

Recent Strategy

- Permissive hypercapnia and hypoxemia
- Pressure-limited ventilation (<25 cmH₂O)
- Postductal paCO₂ 40-55 mmHg
- Preductal SpO₂ 85-90%
- Postductal SpO₂ ignored unless pH is
 < 7.20 or paCO₂ > 65

CONT'D

- Indicated time when the surgical repair is done
 - Reversal of ductal shunting
 - Arterial paCO_2 maintainable < 40 mmHg
 - Hemodynamic stability

Monitoring and Induction

- ASA standard
- NGT to decompress the stomach
- Invasive : arterial line $\pm$ CVP, Adequate IV access
- Foley catheter
- 2 pulse oximeters: preductal and postductal(if $\text{ph} < 7.2$ or $\text{Paco} > 65\text{mmhg}$)
- Precordial stethoscope on the **right axilla**

CONT'D

- Awake intubation
- Rapid sequence IV induction and
- Intubation with assisted or controlled ventilation
- ** avoid mask ventilation or PPV before intubation –iatrogenic lung injury*

Maintenance

- ✓ Volatile agents + IV narcotics + muscle relaxants
- ✓ TIVA
- ✓ Avoid nitrous oxide...??
- ✓ Avoid increase in Pulm.VR leading to R→L shunting:
 - hypoxia, acidosis, hypothermia, pain
- ✓ Treat metabolic acidosis
- ✓ Replace significant blood loss

CONT'D

▪ Mechanical Ventilation

- ✓ adjust FiO₂ to achieve

PaO₂ 80 -100 mmHg

SpO₂ 95 - 98%

- ✓ Small tidal volume to keep airway pressure < 20-30 cm H₂O
- ✓ High respiratory rate 60-120 /min to PaCO₂ 25-30 mm Hg

CONT'D

▪ Potential Problems

Hypoxemia- distension of stomach

1° pulmonary hypoplasia / pulmonary HTN

Contralateral pneumothorax (**sudden desaturation & hypotension**)

Hypotension or IVC compression

Cardiac arrest

Postoperative Care

Ventilatory support

Close fluid management

Hemodynamic monitoring

➤ **“Honeymoon Period”** which may be followed by deterioration:-

increase abdominal pressure

impaired peripheral and visceral perfusion

limited diaphragmatic excursion

worsening of pulmonary compliance

CONT'D

Management (prevent) of Pop PHN:

- ✓ Minimize ETT suctioning
- ✓ Vasodilators : rarely effective

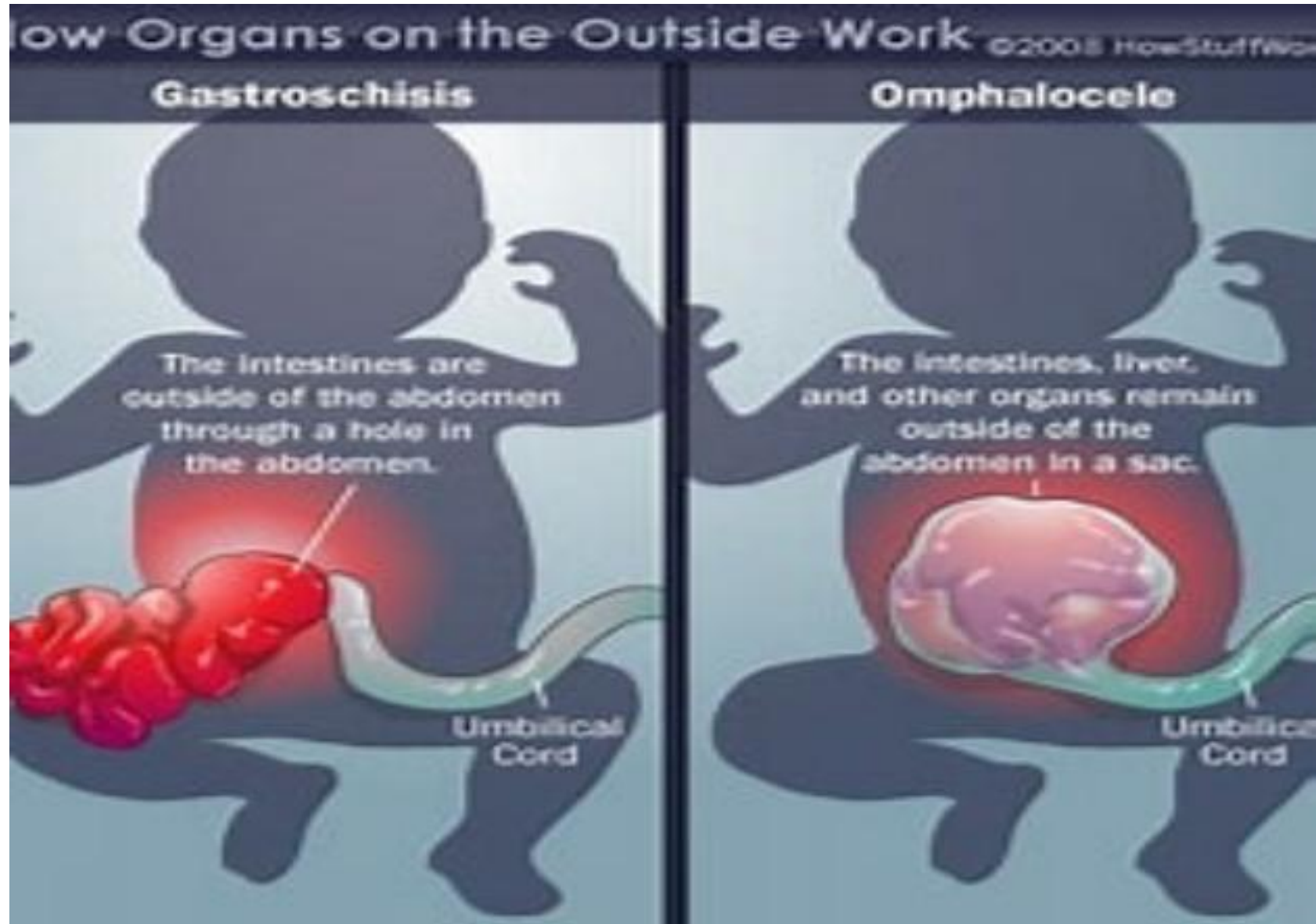
tolazoline isoproterenol PGE1

nitroglycerin SNP

- ✓ Inhaled nitric oxide

Omphalocele and Gastroschisis

- They are major defects in the closure of the abdominal wall that result in exposure of viscera that are either covered (omphalocele) or not covered (gastroschisis)



Omphalocele

➤ Incidence

- 1-2.5 per 5000 live births
- Male preponderance
- Second most common abdominal wall defect

➤ Etiology

- Cause is unknown
- Rare reports occurring in families
- No specific gene has been identified

CONT'D

➤ **Associated Anomalies**

- 30-70% will have associated anomalies
- **Congenital heart disease – up to 45%**
 - VSD /ASD
 - Ectopia Cordis
 - Tricuspid Atresia
 - Coarctation of the Aorta
 - Persistent Pulmonary Hypertension

CONT'D

- **Chromosomal Abnormalities – up to 40%**
- **Beckwith-Wiedemann syndrome**
- **Down's Syndrome**
- **Musculoskeletal abnormalities**
- **Neural tube defects**
- **GERD – up to 43% in one study (cannot be generalized)**
- **Macrosomia**

Key features of BWS

- Overgrowth
- Macrosomia:
- Facial features
- Abdominal wall defects (omphalocele)
- Organomegaly
- Hypoglycemia
- Tumor risk(Wilms tumor or hepatoblastoma).

Pathology omphalocele

- Central abdominal wall defect
- Usually greater than 4cm in diameter
- Covered by a translucent sac unless ruptured inutero
- Muscles of the abdominal wall are normal
- Liver, midgut and spleen and gonads in the sac usually
- All have malrotation
- Infants are usually full term

CONT'D

- **Diagnosis**-Usually detected at 18 +/- 6 weeks
- **Antenatal**- by ultrasound-sensitivity 75%
 - maternal serum α -fetoprotein (AFP)
- **Postnatal**- P/E



Anesthetic Management

- Newborn may be delivered vaginally
- Cardiology evaluation and echo – do not delay surgery
- Surgery
 - Small defects may be closed primarily
 - More often staged procedure with placement of a silo
 - ❖ Its reduced daily if tolerated at bedside– ideally in 7days
 - ❖ Operating room for final closure

Preoperative stabilization

- Management of respiratory insufficiency,
 - Establishment of adequate intravenous access, and
 - Assessment for associated congenital anomalies
-
- ❖ Respiratory failure at birth in this infants is a significant predictor of mortality- **Lung hypoplasia and abnormal thoracic development**
 - ❖ Anticipate difficult AW (**if Beckwith-Wiedemann syndrome**)
 - ❖ Priority for intravenous fluids/Electrolyte imbalances correction
 - ❖ Prophylactic antibiotics

CONT'D

- Small % of children with omphalocele will also have:
 - Beckwith-Wiedemann syndrome
 - Characterized by profound hypoglycemia
 - Hyperviscosity syndrome
 - Congenital heart disease
 - Associated visceromegaly

Intraoperative Care

- Primary repair of the deficit can be done if the defect is small
- Avoid nitrous oxide - **↑ses the volume of gas in the intestine**
- Muscle relaxation is necessary to allow closure of the abdomen.
- Pulse oximeter probe on the foot- **monitoring circulation to the lower extremities during abdominal wall closure**

CONT'D

- The three major perioperative concerns for both omphalocele & gastroschisis are:-
 - ✓ fluid loss,
 - ✓ hypothermia and
 - ✓ ventilation

Post Operative Care

- IV fluids to maintain adequate urine output
- Wean ventilator after final closure
- NPO until return of bowel function - may be delayed
- Remedy can be- TPN

Outcomes

- Post operative complications
 - ✓ Hypovolemia – very high fluid requirements
 - ✓ Tight closure -> ventilatory compromise, ↓sed venous return and low cardiac output, oliguria
 - ✓ Metabolic acidosis (kinking of the hepatic veins)
- Survival rates range from 70-95%
- Mortality is often related to associated anomalies

Gastroschisis

➤ Incidence

- 2-5 per 10,000 live births
- Male preponderance
- **Most common abdominal wall defect**

➤ Etiology

- Cause is unknown, but may be related to fetal distress
- Infants are more likely to be premature
- Term babies are more likely to be SGA

CONT'D

- **Associated Anomalies**

- Much less common than with omphalocele
- Intestinal atresia may be seen in 10-15%
- GERD ~ 16%
- Undescended testicle ~15%
- Malrotation is always present

CONT'D



Pathology

- Small abdominal wall defect: usually 2-4cm
- Occurs adjacent to the umbilicus (mainly to the right)
- Abdominal muscles are normal
- There is no sac
- Mid gut is herniated through the defect
- Matted bowel loops may be present at birth
- The exposed bowel quickly becomes matted with fibrinous exudate such that individual loops can not be distinguished

Diagnosis

- Antenatally by ultrasound-Sensitivity reported to be 83%
- Detected at 20 weeks +/- 7 weeks
- Maternal serum α -fetoprotein (AFP)

Anesthetic Management

Almost the same as **omphalocele** , but focus to

- Cover exposed bowel with warm saline moistened sterile towel
- Support of the premature infant
- Monitor for fluid and heat loss
- Should be taken to the operating room
- Faster bowel reduced, more likely be primary closure done
- If not achieved, silo is placed
- Post operatively requires adequate fluid resuscitation

Post operatively

- Wean ventilator after final closure
- NPO until return of bowel function
 - May be more prolonged than with omphalocele
- TPN



Prognosis

- Post operative complications
 - Hypovolemia
 - Closure too tight -> ventilatory compromise, decreased venous return and low cardiac output, oliguria
- Survival rate is ~90%
- Complications related to prematurity responsible for most morbidity and mortality

CONT'D

- Primary fascial closure has been reported in up to 80%;
some recommend silos in all cases
- Median time to feeding ~30 days (5-60 days reported range)
- Atresias are repaired several weeks after intestinal reduction
and fascial closure

Cleft Lip and Palate

- These congenital anomaly are relatively common
- About 1/3rd are associated with other syndromes
- They need a careful and thorough pre-operative asses't
- Cleft lip is usually repaired at 3 to 6 months of age
- Cleft palate is repaired at 9 to 12 months of age
- Difficult airway may occur but is unusual unless there is also retrognathia

Pierre-Robin Sequence



- Micrognathia
- Glossoptosis
- U-shaped Cleft palate
- Associated cardiac abnormalities-
pulmonary stenosis
- Airway Problem
 - Difficult mask and intubation



“U” shaped cleft palate



“V” shaped cleft palate

CONT'D

- In children with large or bilateral clefts, the tongue may impinge in the cleft obstructing the airway or
- The laryngoscope blade may fall into the cleft
- Correction of the lip is usually a straightforward procedure
- Cleft palate repair may be complicated by postoperative obstruction
- The child should be extubated awake

CONT'D

- Postoperative analgesia is based on regular PCM and
- Judicious use of opioids
- Infraorbital nerve blocks also provide effective analgesia for cleft lip repair

Inhaled Foreign Body (IFB)

- IFB is a major source of morbidity and mortality in toddlers
- The AWO may be acute, causing ARD and require urgent management; however, often the diagnosis is delayed
- Partial AWO may be undiagnosed for days or even weeks
- The acute choking episode may not have been witnessed and the child may present late with signs of pneumonia

CONT'D

- The urgency of removal depends on:
 - The degree of respiratory symptoms
 - The likely location of the obstruction
- Ideally the child should be fasting
- Clear, effective, and ongoing communication with OR team
- Anti-cholinergic and steroids are given prior to induction

CONT'D

- Ideally, SV is maintained (reduce the risk of positive ventilation pushing the object further distally, which makes extraction more technically challenging)
- After deep anesthesia, LA should be directly applied to the glottis and trachea
- Maintenance can be with IAAs through the bronchoscope or with a propofol infusion

CONT'D

- Propofol infusion provides a reliable delivery and avoids the issue of volatile agent exposure to the surgeon
- Remifentanyl infusion (may obtund AW reflexes) but
- Should be used judiciously to avoid unwanted apnea
- Muscle relaxation may be required in some situations
- It shouldn't be given before establishing that positive pressure ventilation is safe

Acute epiglottitis

- Is a bacterial infection (*Haemophilus influenzae* type B) classically affecting 2- to 6-year-old children
- It may rapidly progress from a sore throat to dysphagia and complete airway obstruction
- The inflammation typically involves all supraglottic structures
- They usually prefer a sitting position for their airway patency

CONT'D

- ETT intubation and antibiotic therapy can be lifesaving
- Epiglottitis has increasingly become a disease of adults
- Children with impending airway obstruction from epiglottitis present in the OR for definitive diagnosis by laryngoscopy followed by intubation

CONT'D

- A preoperative lateral neck X-ray show a characteristic thumblike epiglottic shadow(very specific but often absent)
- It may be helpful in showing other cause of obstruction like FB
- Stridor, drooling, hoarseness, rapid onset and progression, tachypnea, chest retractions, and a preference for the upright position are predictive of airway obstruction in these children

CONT'D

- Total obstruction can occur at any moment
- Adequate preparations for a possible tracheostomy must be made prior to GA induction
- No laryngoscopy before anesthesia induction(laryngospasm)
- In most cases, IAA induction is done with patient in the sitting position using an IAAs and a high concentration of oxygen

CONT'D

- Oral intubation with an ETT 1/2 to 1 size smaller than usual is attempted just after adequate depth of anesthesia is established
- The oral tube may be replaced with a well-secured nasal endotracheal tube at the end of the procedure
- The latter is better tolerated in the postoperative period
- If intubation is impossible, rigid bronchoscopy or emergency tracheostomy must be performed

A case discussion on anorectal malformation

- A 2-day-old male newborn has not passed any meconium since birth.

Examination of the newborn reveals no anal orifice. The newborn is scheduled for a colostomy.

- What is your pre-operative mgt plan?
- How would you manage anesthesia?
- What are your post-operative considerations?

PREOPERATIVE CONSIDERATIONS

- An imperforate anus may occur with many presentations
- Type 1: Anal stenosis occurs at the anus or just above the anus
- There may be an associated fistula with the genitourinary system or the perineum
- Type 2: Imperforate anus occurs when the obstruction is due to a persistent membrane

CONT'D

- Type 3 (the most common): Imperforate anus is associated with the rectum ending blindly distant above the imperforate anus
- The rectum may have a fistulous connection to the urethra, bladder, base of the penis or scrotum, or vagina
- Type 4: There is a normal anus and anal pouch, but the rectum ends blindly

CONT'D

- The newborn may present with a distended abdomen
- The newborn may pass stool (near vaginal opening, base of the scrotum, the bladder, or the penis)
- Urinary tract infection should be treated before surgery
- It may also occur with other congenital abnormalities (VACTERL) syndromes
- The higher the anorectal anomaly is located, the greater the incidence of associated anomalies

CONT'D

- Its incidence is about 1:5000 newborns
- The newborn should be screened for associated abnormalities preoperatively, especially for cardiac and tracheoesophageal anomalies
- The plan of pain mgt also should be revealed preoperatively
- The child may be on a colostomy active or stenosed
- Basic lab. tests and their respective mgt

ANESTHETIC MANAGEMENT

- Select the anesthetic technique depending on:
- The extent of the abdominal distention
- The type of operation (simple perineal anoplasty, temporary colostomy, or extensive abdominoperineal repair), and
- Whether there are any associated CVS and Rs problems
- Consider rapid-sequence induction or awake intubation if the abdomen is distended

CONT'D

- Anesthesia can be maintained with an IAAs in an air-oxygen mixture
- NMBDs may allow for better operating conditions
- If unable to identify the anal sphincter (use narcotics judiciously)
- Nitrous oxide should be avoided to prevent further bowel distention

CONT'D

- Ensure adequate hydration
 - Large insensible fluid losses
 - Large third space losses with bowel manipulation
 - An additional considerations if there is a bowel perforation
- Balanced salt solutions or colloid at a rate of 10 mL/kg/h of surgery or more may be needed

CONT'D

- Consider the risk of sepsis if there was bowel perforation
- Monitor urine output, blood pressure, and quality of heart tones
- Invasive monitoring is not required for the healthy patient

POST-OPERATIVE CONSIDERATIONS

- If there are other associated life-threatening anomalies
- If a large volume of fluids was required to keep the patient hemodynamically stable
- Strict post-operative care is required

Tonsillectomy and Adenoidectomy

- Lymphoid hyperplasia can lead to UAWO, obligate mouth breathing, and even pulmonary hypertension with cor-pulmonale
- All children undergoing tonsillectomy or adenoidectomy should be considered to be at increased risk for perioperative airway problems

Anesthetic Considerations

- Postpone surgery for evidence of acute infection or suspicion of a clotting dysfunction (Recent aspirin ingestion)
- Pre-op anticholinergics will decrease pharyngeal secretions
- A history of AWO or apnea suggests IAAs induction without paralysis until ventilation with positive pressure is secured
- A reinforced or preformed ETT(RAE tube) may decrease the risk of kinking by the surgical retractor

CONT'D

- Consider for an occult blood loss
- A gentle inspection and suctioning before extubation
- Deep extubation decreases the chance of laryngospasm and may prevent blood clot dislodgment from coughing
- Awake decreases the risks of aspiration (experts prefer)

POST OPERATIVE CONCERNS

- Postoperative vomiting is common
- Postoperative bleeding in the recovery room
- Evidenced by restlessness, pallor, tachycardia, or hypotension
- If reoperation is necessary to control bleeding, intravascular volume must first be restored

CONT'D

- Gastric emptying with NG tube is followed by a RSI with a gentle cricoid pressure
- There may be bleeding and AWO, children younger than 3-years may be admitted for the first Post-op night
- Sleep apnea and recent infection increase the risk of postoperative complications

Trisomy 21 Syndrome (Down Syndrome)

- A third chromosome 21 results in the most common congenital pattern of human malformation
- Characteristic abnormalities of interest are a **short neck, irregular dentition, mental retardation, hypotonia, and a large tongue**
- Associated abnormalities include CHD in 40% of patients (endocardial and ventricular septal defect), **subglottic stenosis**, TEF, chronic pulmonary infections, and seizures

CONT'D

- These neonates are often premature and SGA
- Later in life many patients with Down syndrome often require multiple procedures requiring general anesthesia

Anesthetic Considerations

- These patients often have difficult airways (mainly at infancy)
- Airway Problem
 - May be difficult bag mask ventilation from macroglossia
 - A-A instability- need to maintain in-line neck stabilization
 - Potential subglottic stenosis
- One size smaller ETT than that predicted by age

CONT'D

- Postoperative stridor and apnea are common(follow up is needed)
- Careful neck flexion during laryngoscopy and intubation
- Risk of atlantooccipital dislocation (laxity of these ligaments)

CONT'D

- Always consider a risk of associated congenital diseases
- Avoid air bubbles in the intravenous line because of possible right-to-left shunts (paradoxical air embolus)
- Anesthetic Concerns
 - 40-60% are born with cardiac anomalies
 - Large tongue, small mouth

CONT'D

- Commonly have adenotonsillar hypertrophy
- Atlanto-Axial instability
- Bradyarrhythmias
- ↑sed risk for subglottic stenosis, tracheomalacia/ narrow glottis
- Behavioral issues (special need population)
- Potentially difficult IV access

SPINA BIFIDA

CONT'D

- Spina bifida is a diverse group of congenital malformations of the spine and spinal cord
- These conditions result from a **neural tube fusion failure**
- Malformation can be an open defect with exposed neural tissue such as **myelomeningocele and myeloschisis** or
- A closed defects with a skin covering such as **meningocele, tethered cord, and split cord**

CONT'D

- Open defects are associated with a neurologic deficit
- Are almost always repaired perinatally (fetal surgery)
- Closed defects may have associated deficit or may be asymptomatic and undiagnosed until adulthood

CONT'D

- Neurologic abnormalities include
 - Motor and sensory deficits
 - Chiari II malformation
 - Hydrocephalus
 - Neurogenic bladder
- These patients may present for orthopedic surgery for:
 - Correction of congenital or acquired limb deformities
 - Such as clubfoot or hip dislocation
 - Spinal fusion for scoliosis, or release of contractures

Anesthetic Implications

- Repeated surgical exposures (increased latex allergy)
- Challenging neuraxial anesthesia
- Whether or not they have had corrective spinal surgery,
- Anatomic abnormalities in the spine (risk of dural puncture), failed block, and neurologic injury
- Malformed or absent ligamentum flavum (loss of resistance technique for epidural catheter placement ?)

CONT'D

- The epidural space may be abnormal/ nonexistent in who had prior surgical repair
- Cord tethering can result in low termination of the spinal cord and posterior placement of neural elements

CONT'D

- If neuraxial anesthesia is indicated in these patients
- For detail understanding of spinal structures **obtain MRI:**
 - For examination of bony and ligamentous defects
 - The level of termination of the spinal cord, and
 - The presence of masses such as lipoma

CONT'D

- Avoid needle placement through a surgical scars
- Epidurals should be placed above the spinal defect
- Smaller than usual epidural boluses are recommended
- Failed or incomplete block may result in the need for rescue analgesia or conversion to general anesthesia

Meningomyelocele (MMC)

- The hernial protrusion of a part of the meninges and spinal cord through a defect in the vertebral column
- 2^o to lack of maternal folic acid intake and antenatal screening

CONT'D

- Consider the following in addition to usual neonatal mgt
 - Special positioning for tracheal intubation
 - Possibility of underestimating fluid and blood loss
 - High association of MMC with hydrocephalus
 - Possibility of cranial nerve palsy (**inspiratory stridor**)
 - Potential for brainstem herniation
 - The valsalva maneuver and effective communication with surgical team

CONT'D

- The anesthetist must establish good IV line to replace all fluid deficits, including loss from the defect (with N/S),
- Ensure that cross-matched blood is available
- Latex allergy precautions should be used with these children for their first and all subsequent anesthetics

ANAESTHESIA FOR VENTRICULO-PERITONEAL SHUNT INSERTION

Brainstorming

- What are the symptoms and signs of raised ICP?
- What are the causes of hydrocephalus in children?
- What mechanisms regulate cerebral blood flow?
- What is the relationship between ICP and CPP and how can we manipulate this therapeutically?
- What additional considerations are there if the patient is a neonate?

INTRODUCTION

- An anaesthetic techniques employed for the insertion of ventriculoperitoneal (VP) shunts
- Basic principles of neurophysiology to treat patients with raised intracranial pressure(ICP)
- Shunting started as a treatment for ydrocephalus in the 1940s
- These children affected had a poor prognosis before

CONT'D

- Only 20% of those who did not receive surgery reached adulthood and
- About 50% of the survivors were brain damaged (without surgery)
- The development of valved and then silicone shunt systems dramatically improved their prognosis

ICP AND CPP

- The management of patients undergoing VP shunt insertion is dependent on understanding:
 - The relationship between ICP and CPP and
 - How this is affected by any disease process, anaesthetics and surgery
- The cranial contents consist of
 - The brain tissue (80%)
 - The blood (10%) and
 - The cerebrospinal fluid (10%)

CONT'D

- Volume of the cranial vault is fixed after 2-years (cranial sutures fuse)
- In neonates, volume of the vault can expand if intracranial contents increase gradually (e.g. congenital hydrocephalus)
- This expansion has a peak when reached will lead to $\uparrow$ in ICP
- Any changes in the volume of one cranial contents can lead to an alteration in ICP
- This has both pathological and therapeutic implications

CONT'D

- CBF determines CBV and is higher in children than adults (100 vs 50 ml/100g/min)
- It is coupled to the metabolic demands of the brain and regulated via oxygen requirement, PaCO₂ and intracerebral acidosis

CONT'D

- Autoregulation (relatively constant BF across a range of arterial pressures, as low as 40 mmHg in small children)
- This is not available in prematures or during pathological processes (e.g. infection)
- The relationship b/n CBF and arterial pressure is more linear

CONT'D

- ↓sing PaCO₂ causes cerebral vasoconstriction, ↓sing CBF and hence CBV and ICP
- This is at the expense of oxygen delivery
- Therefore hyperventilation should be used as a short-term measure when ICP is very high (before definitive treatment)

The Cerebrospinal fluid (CSF)

- The CSF is continuously produced by choroid plexus, and
- Absorbed at the arachnoid villi after circulating through ventricles
- In children the rate of production of CSF is 0.2-0.4 ml/min with around 250 ml produced and absorbed /day
- At any given time around 70mls is present in the head
- Interruption to normal flow, ↑sed production or ↓sed reabsorption of CSF can result in hydrocephalus which may lead to raised ICP

CONT'D

- The classical distinction b/n obstructive and communicating hydrocephalus is less useful clinically
- The cause of reduced absorption of CSF in communicating hydrocephalus is usually functional obstruction at the arachnoid villi (e.g. by blood or protein)

CONT'D

- Brain tissue may be pathologically enlarged by any space occupying lesion which may lead to an ↑ in ICP
- CPP is defined as the difference between the MAP and the sum of ICP and the central venous pressure (CVP)
- $CPP = MAP - (ICP + CVP)$
- Normal range of ICP in neonates is 2-4 mmHg (7-15 mmHg in adults)

CONT'D

- ↑sed ICP gives rise to such symptoms and signs as:
 - vomiting, irritability, drowsiness, bulging fontanelle (if still present), ↑sed head circumference, downward gaze of the eyes (“setting sun” sign), Cushing’s response
- At ICP > 20mmHg focal ischaemia occurs and at >50 mmHg global ischaemia occurs if CPP remains constant
- This is the reason MAP is raised therapeutically with the use of vasoconstrictors in some cases of raised ICP

CONT'D

- Long term ↑ses in ICP can impair neurological development
- The acute rise in ICP causes distortion and displacement of the cerebral contents through the foramen magnum leading to coma, respiratory arrest and death
- Most anaesthetic agents cause a reduction in MAP but this potentially deleterious effect is usually offset by the corresponding reduction in the CMRO₂

INDICATIONS FOR SHUNTING

- The causes of hydrocephalus may necessitate a VP shunt insertion
- **Congenital :**
- Myelomeningocele
- Stenosis of the cerebral aqueduct
- Arnold-Chiari malformation, Arachnoid cysts
- Vascular malformations
- Idiopathic

CONT'D

Acquired :

- •Infection
- Intraventricular haemorrhage
- Tumour
- Trauma
- Nutritional deficiencies

CONT'D

- VP shunts are done for a range of patients (newborns to older children may be with a number of potential co-morbidities)
- A shunt provides a route for the CSF to drain and ↓ ICP
- The proximal end is placed in the ventricle and the distal end at a site where the CSF can be absorbed (peritoneum but atria and pleura are also used)

CONT'D

- The complications associated with VP shunts are :
 - Shunt blockage, infection and over drainage, all of which may necessitate revision surgery
- Endoscopic third ventriculostomy (ETV) is an alternative treatment for hydrocephalus in carefully selected patients
- A neuroendoscope is used to make a hole in the floor of the third ventricle to facilitate CSF drainage

CONT'D

- The technique was originally developed in the 1900s
- Abandoned due to severe complications likely attributable to poor anaesthesia and surgical equipments
- Developments in optics, image processing and improvements in anaesthesia have enabled ETV to be used safely with comparable results to VP shunts (Warf et al have demonstrated in their series from Uganda)

CONT'D

- ETV has benefit of avoiding long-term VP shunt complications &
- The attendant costs of follow up and revision surgery, which has enormous implications for developing countries
- The incidence of infantile hydrocephalus is about 3-5 cases per 1000 live births
- The peak ages of presentation include the first weeks of life, 4-8 years and early adulthood

CONT'D

- An estimated 750,000 people have hydrocephalus, approximately 160,000 VP shunts are inserted worldwide each year
- Over 56,600 children under 18-yrs thought to have a shunt placed
- Depending on the indication for VP shunt insertion, the procedure may be urgent or an emergency
- With an additional attendant risks (full-stomach, abnormal electrolytes, hypovolaemia, precipitously high ICP)

PREOPERATIVE ASSESSMENT

- An essential component of any anaesthetic technique
 - Taking anaesthetic history of problems with previous anaesthetics
 - Family history of reactions to anaesthesia
 - Past medical history
 - Current medication (e.g. anticonvulsants, acetazolamide, furosemide)
 - Allergies and last oral intake

CONT'D

- Particularly give attention for possible co-morbidities
- Cardiovascular (e.g. congenital cardiac disease) and
- Respiratory systems (e.g. bronchopulmonary dysplasia, kyphoscoliosis, recurrent respiratory infections secondary to neurological dysfunction)

CONT'D

- Assessment of airway, cardiorespiratory and neurological systems, which in cases of raised ICP may demonstrate reduced LOC
- With high risk of pulmonary aspiration
- Volume status should be checked (vomiting and dehydration) may necessitate intravenous fluid preoperatively

CONT'D

- Review CBC to certain the level of Hgb and clotting function so that appropriate blood products can be ordered as indicated
- Urea and electrolytes should be checked if indicated
- Sedative premedication should be used carefully as it may exacerbate or mask signs of neurological dysfunction, an anxious combative patient may have a rise in ICP during induction

CONT'D

- The patient should be carefully monitored
- If the preop workup reveals a multiple co-morbidities, such as an ex-premature infant with chronic lung disease
- The need for postop additional respiratory support
- Should be considered and organised in advance

INDUCTION OF ANAESTHESIA

- Before induction all anaesthetic equipments and drugs should be checked to ensure that they are in a good working order
- A means of oxygen delivery, oxygen and volatile anaesthetic agent, suction, a tilting trolley and monitoring should be ready
- Different size facemasks, OPA, ETTs and laryngoscopes
- An appropriate breathing circuit should be checked for correct configuration and patency

CONT'D

- Standard monitoring should include pulse oximetry, electrocardiogram, non-invasive blood pressure, oxygen concentration, and capnography
- Then, a method of induction is determined by the circumstances of the case and the preference of the patient and the anaesthetist

CONT'D

- IV induction allows for rapid control of the airway in emergency situations
- The patient is not fasted and difficult AW is not anticipated
- This may be achieved using a suitable induction
- Propofol 2-4 mg/kg, thiopentone 3-5 mg/kg)
- Avoid hypotension, CPP may decrease in the face of raised ICP
- Ketamine should not be used as it can increase ICP

CONT'D

- Muscle relaxation can better achieved with NDNMDs
- Succ can be used if the risk of aspiration outweighs the problems of transient increases in ICP
- Gaseous induction is an alternative (with a non-irritant IAAs such as sevoflurane or halothane)
- A rapid gaseous induction is preferable to prolonged attempts at cannulation in a distressed and unwell child

CONT'D

- Once anaesthetized, IV should be established quickly to use of a NMB to facilitate airway control
- Avoid Hypercarbia (cerebral vasodilation and worsen raised ICP especially with laryngoscopy and airway manipulation)
- Secure airway by tracheal intubation with appropriate sized ETT
- This should be a reinforced (armoured) tube if available
- The standard tube can be used with effective communication with surgical team and frequent inspection

CONT'D

- The tube should be firmly secured with waterproof tape and dressing and the eyes should be padded and taped
- The patient should be positioned to allow good access for the surgeon and to avoid any undue pressure on vulnerable areas
- Consider to optimizing venous drainage to increase CPP, reduce venous bleeding and improve the surgical field
- Monitor core temperature (rectal or oesophageal) and warming devices (e.g. warm air blankets) to maintain normothermia

MAINTENANCE OF ANAESTHESIA

- Maintenance can be with a volatile agent and oxygen
- Goal to maintain CPP until the raised ICP is relieved by positioning of the VP shunt
- So, avoid hypotension and control MV to maintain normocarbica (ET CO₂ 4-4.5 kPa)
- This optimises CPP and avoids increases in ICP

CONT'D

- Minimize PEEP to avoid venous congestion in the head but may be used if there are difficulties in maintaining oxygenation
- The most stimulating parts of the surgery is the initial incision and tunnelling under the skin
- Fentanyl 1-3 mcg/kg or increased depth of anaesthesia, can be used to prevent HR increase and ICP

CONT'D

- Post-operative analgesia can be provided by:
- A combination of LA infiltration like bupivacaine 0.25% (0.5-0.75 ml/kg) and PCM 15 mg/kg IV or PR
- Avoid high doses of long acting opioids as this may have a potential detrimental effects on conscious level

POSTOPERATIVE MANAGEMENT

- At the end of surgery, reverse the NMBDs by using neostigmine=50 mcg/kg IV with atropine= 25 mcg/kg IV
- Extubation can be done once awake, avoiding hypercarbia, and with a technique of low risk of aspiration (lateral or sitting)
- Achieve analgesia with a combination of PCM and NSAIDs, if not contraindicated add oral opioid for breakthrough pain

CONT'D

- The patient should be monitored in a suitable environment by those familiar with paediatric neurosurgical cases and able to carry out regular neurological observations
- e.g. AVPU than GCS

THE NEONATAL PATIENT

- Many patients requiring VP shunt insertion are neonates
- They are physiologically immature and so present additional considerations for the anaesthetist
- Many are premature with a LBW, with congenital hydrocephalus needing urgent surgical treatment in the first days of life

CONT'D

- They are most likely to have complications with their VP shunts and require frequent surgery to correct these problems

Pre-operative evaluation

- Look for problems associated with neonates and prematurity such as anaemia, coagulopathy, jaundice, respiratory disease and congenital cardiac problems
- Ixs like echocardiogram may be indicated
- Blood and other required products should be readily available

INDUCTION AND MAINTAINANCE

- Induction with sevoflurane provides good CVS stability
- Neonates are a more technically challenging group
- Particular attention must be paid to securing ETT and IV cannulae as endobronchial intubation and extravasation are commonly encountered problems

CONT'D

- Congenital hydrocephalus may distort anatomy of skull which can make AW management more difficult
- A pillow placed under body of neonates can facilitate laryngoscopy
- The use of a naso-gastric tube is an important component of managing the neonatal airway and ventilation
- Handling should be careful and minimal to reduce the risks of complications

CONT'D

- They have a high surface area to volume ratio and so lose heat quickly
- They have less developed mechanisms for thermal homeostasis
- Core temperature must be recorded and active measures taken to maintain body heat

CONT'D

- These may include:
 - Increasing the ambient temperature
 - Warm air blankets
 - Overhead heaters
 - Heat-moisture exchangers to warm inspired gases
 - Warming intravenous fluids and
 - Covering of exposed areas

CONT'D

- Metabolism is different as they have immature liver but a relatively higher volume of distribution
- Paracetamol, local anaesthetics, volatile anaesthetic agents and opioids are all used in relatively lower doses
- Fluid balance as renal function is immature in the neonate
- Infusion of 10% glucose may be required to maintain blood glucose within the normal range

CONT'D

- Regular monitoring of blood glucose and electrolytes allows for adjustment of fluid regimens as required

POST OPERATIVE CARE

- Some neonates will require further respiratory support (especially pre-term infants with bronchopulmonary dysplasia)
- Patients that can be extubated will need apnoea monitoring for 24-hour hours postoperatively
- They also need regular neurological observations

Monitor pediatric patients

- Pediatric monitoring is similar to adults in many aspects
- Monitor pediatric patients continuously with precordial or esophageal stethoscope
- Which will allow you to detect changes in the rate, quality, and intensity of the heart sounds
- Pulse oximetry, ETCO₂, BP(fitted sized cuffs), temperature, and ECG should also be monitored routinely

CONT'D

- More invasive or sophisticated monitoring should be reserved or indicated procedures
- With children with CHD there is increased gradient b/n ETCO_2 and PaCO_2 level (V/Q mismatch)
- In cyanotic CHD, ETco_2 underestimates Paco_2 as venous blood passes directly into the arterial circulation without going through the lungs

CONT'D

- Low VT, rapid RR, and changing intrapulmonary shunts make ETco₂ inaccurate for infants and premature neonates with respiratory distress syndrome
- **Transesophageal echo** is important intraoperative assessment tool for cardiac function, flow defects, cardiac morphology, and adequacy of repair during and after surgical procedures for CHD
- Use of PA catheters is quite limited in the pediatric population

PAIN MANAGEMENT AND REGIONAL ANESTHESIA

- Neonates, infants, and children experience pain like adults
- Evidences show that neonates have considerable maturation of pain transmission by 26 weeks of gestation and respond to injury with specific behaviors:-
 - Autonomic, hormonal, and metabolic signs of stress and distress
- Extreme pain experienced during neonatal period may have life longlasting adverse effects

CONT'D

- Pain in children can be assessed by self-reporting, observational, and physiological assessment tools
- Self-reporting pain is considered the gold standard, but this assessment method cannot be used in infants, younger children, and in children who suffer from developmental delay
- Children older than 4-6 yrs can self-report pain using face scale

CONT'D

- Evidences indicate that pain in children is underestimated by health care professionals
- Children often receive subtherapeutic doses of analgesics
- The issue of undermedication appears to be particularly problematic in the postoperative setting
- Early painful procedures have been associated with changes in sensitivity to later medical procedures

CONT'D

- PCM continues to be the most common oral analgesic used in children
- It is shown to be safe and effective in neonates older children
- Doses = 10 to 15 mg/kg orally every 4 hours or
- 30 to 40 mg/kg rectally as a loading dose
- Plasma conc. effective for fever and analgesia are 10 to 20 µg/mL

CONT'D

- Ketorolac has become an effective and safe analgesic for pediatric patients
- It can given 0.75 mg/kg IM or with a loading dose of 1 mg/kg IV followed by a maintenance dose of 0.5 mg/kg every 6 hours
- Its disadvantage is prolonging bleeding time,
- Like other NSAIDs, it should be avoided in patients with pre-existing nephropathy or bleeding diathesis

CONT'D

- Ibuprofen most common NSAID given orally to children
- It comes in several palatable preparations
- When given in the recommended oral dose of 10 mg/kg it has similar analgesic effects as acetaminophen or ketorolac
- Gastrointestinal side effects are uncommon

CONT'D

- The safety and efficacy of patient-controlled analgesia (PCA) for children older than 6 yrs has been shown
- While routinely used in children's hospitals
- This is best used only by highly trained medical personal who are knowledgeable in pediatric pain management
- Nurse controlled analgesia for infants and children younger than 6yr
- PCA can be acceptable for palliative care but for posoperative care

Regional Anesthesia

- Because of developmental and cognitive issues, regional techniques are rarely used as a sole anesthetic
- They most often used as adjuncts to general anesthesia and providing postoperative analgesia
- RA techniques (e.g, SA, EA) may also be used as the sole anesthetic in premature infants at risk for postoperative apnea

CONT'D

- Simple techniques such as ilioinguinal—iliohypogastric nerve block, ring block of the penis, or caudal block, local infiltration are useful for pediatric surgical procedures
- **Use of U/S** allows a lower dose/volume of LA to be used
- Particularly important in infants and children of LBW

CONT'D

- **Regional nerve blocks** and direct local infiltration of surgical wounds with long-acting local anesthetics are simple yet very effective and safe methods of providing pain relief for all children

CONT'D

- **Caudal block** is the most commonly used form of RA in children
- This technique can provide postoperative analgesia
- For lower abdominal and genitourinary surgical procedures
- For single-dose administration (outpatient surgeries) bupivacaine, 0.25 - 0.175% solution at a dose of 1 mg/kg is commonly used

CONT'D

- Postop. analgesia typically lasts 4 to 6 hrs and
- This dose is not associated with a motor paralysis
- This route can be used for either a single-dose injection or for catheter advancement for continuous infusion

CONT'D

- **Spinal Anesthesia(SA)** may be used for procedures involving surgical dermatomes below T6
- Note that the dural sac migrates cephalad during the first year of life and in a neonate it is at **S3**
- At age of 1 year it is at the **S1 level**

CONT'D

- The sitting position may be especially helpful in neonates
- It maintains midline needle position and free flow of CSF
- SA is a particular good option for **premature infants** who undergo surgery as the incidence of postoperative apnea has been shown to be reduced in these infants with the use of this technique

Questions?

